

Cognitive Load Across Common VR Locomotion Techniques

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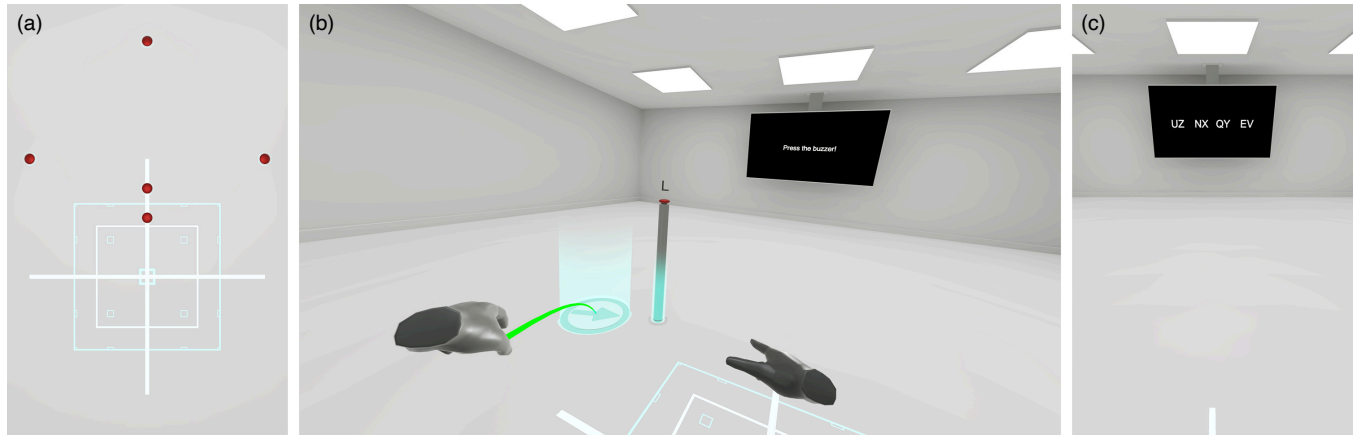


Figure 1: Overview of the experimental setup and task elements. (a) Top-down view of the virtual environment showing the predefined buzzer locations; each location is used twice per locomotion method. Rectangles marked with a cross denote the starting position for post-activation return. (b) Teleportation locomotion method, illustrating the teleport indicator and the letter cue displayed above the buzzer, which specifies the required activation hand. (c) Virtual display presenting the letter pairs shown to participants during the memorization phase.

Abstract

Virtual reality applications rely on diverse locomotion techniques, yet their cognitive demands remain insufficiently understood. This study investigates how three widely used VR locomotion methods—*Real*, *Joystick*, and *Teleport*—affect cognitive load in both standing and sitting postures. In a mixed-design experiment ($N = 42$), participants completed a dual-task combining spatial navigation with working memory requirements. *Joystick* consistently yielded superior performance and the lowest subjective workload, whereas *Teleport* with orientation control imposed the highest cognitive demands. These findings highlight that implementation details such as orientation control and input modality critically influence the cognitive load of VR locomotion and provide actionable insights for designing more effective VR experiences.

Keywords

Cognitive Load, Virtual Reality, Locomotion Techniques, Dual-Task Performance, Working Memory, Mixed-Design Experiment, Posture Effects, Cognitive Performance

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1 Introduction and Related Work

Cognitive Load Theory states that human attentional resources are limited and that processing extraneous, non-task-related information competes with task-relevant processing, thereby reducing cognitive performance and learning outcomes [6, 12, 23, 24]. As virtual reality (VR) is increasingly used across application domains such as training, education, and entertainment [10, 15, 18–20, 22], managing cognitive load becomes particularly important: excessive mental demands can impair user performance, learning, and the overall effectiveness of VR experiences [1, 14, 25].

Previous work has examined cognitive load in VR primarily using dual-task paradigms that combine spatial tasks with memory or motor demands [13, 17, 21]. Such paradigms typically assign a memory task such as letter recall as the primary task, while adding a secondary spatial task such as mental rotation to increase cognitive load. A range of factors has been shown to influence cognitive load, including environmental complexity [16], the presence of bystanders [17], self-avatar representation [13, 21], level of immersion [11], and gestural interaction [4, 5].



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Lai *et al.* [9] investigated the impact of VR locomotion techniques on cognitive load in two within-subjects studies. They found that target-based techniques were more efficient and induced lower cognitive load than steering-based techniques. In the target-based conditions (e.g., teleportation), participants specified only a target position; their resulting view orientation was neither defined nor controlled. In contrast, steering-based techniques required continuous joystick input for movement while viewing direction remained freely controlled by head tracking, resulting in higher cognitive demands.

Hořejší *et al.* [7] found that joystick locomotion provided the best balance between navigation speed and low cybersickness compared to hand-based teleportation and Cybershoes. Kazemi *et al.* [8] further showed that standing users experience higher mental workload during teleportation than seated users; however, their study compared only teleportation and joystick locomotion and did not examine additional locomotion techniques such as natural walking.

With modern standalone head-mounted displays, several locomotion techniques have become widely adopted [2, 3, 26]. These include natural walking (*Real*), in which users physically move within the real space; controller-based joystick locomotion (*Joystick*); and teleportation with optional orientation adjustment (*Teleport*). All three techniques are provided as standard components in major VR development environments such as the Unity XR Interaction Toolkit, Unreal Engine VR Template, and OpenXR, making them prevalent in both research and applied VR development.

To the best of our knowledge, no study has systematically examined how these three locomotion techniques differ in cognitive load across *standing* and *sitting* postures, despite their widespread use. We address this gap through a 2×3 mixed factorial design, investigating the combined effects of posture and locomotion technique on cognitive performance and subjective workload.

2 Method

2.1 Locomotion Methods

Real. Participants navigate the virtual environment through natural physical walking, with real-world movements mapped directly to six-degrees-of-freedom motion in VR. The virtual scene continuously updates based on the participant's tracked position and orientation. In the seated posture condition, participants sit on a wheeled office chair and propel themselves through the physical space using their feet, enabling continuous physical locomotion while seated.

Joystick. Participants use the VR controller thumbsticks for navigation. Translational movement is controlled via the left thumbstick, with movement speed proportional to the degree of deflection. Rotational movement is controlled via the right thumbstick; a full deflection to either side produces a discrete 45-degree snap rotation in the corresponding direction. Successive deflections allow participants to perform multiple consecutive snap rotations.

Teleport. Participants navigate the virtual environment using a teleportation technique. Touching the left controller's trigger displays a teleport indicator at the target location. By default, the indicator's facing direction matches the participant's current view

orientation. Participants can adjust their desired post-teleportation orientation using either thumbstick, which rotates a directional arrow indicating the resulting view direction (see Figure 1). Upon fully pressing and releasing the trigger, participants are teleported to the selected target position with the specified orientation.

2.2 Cognitive Task

Previous studies [13, 16, 17] commonly used mental rotation as a secondary task alongside letter recall to assess cognitive performance. However, as mental rotation is typically used to increase the difficulty of letter recall but does not fit our interaction requirements, we design a study-specific cognitive task inspired by this paradigm.

Our task imposes concurrent demands on working memory, spatial attention, and motor coordination to reflect the cognitive workload typically experienced during locomotion in virtual environments. The experiment consists of repeated trials, each following the same sequence of subtasks. Each trial begins with the presentation of four letter pairs on a virtual monitor for 15 seconds, which participants are instructed to memorize.

After this memorization phase, a buzzer appears at an unpredictable, yet predefined location (see Figure 1) within the virtual environment to ensure comparability across trials. Participants are required to reach the buzzer as quickly as possible using the assigned locomotion method and to activate it with the designated hand. As illustrated in Figure 1, a letter cue ("L" or "R") above the buzzer indicates the required activation hand. This cue continuously rotates to face the participant's field of view, ensuring clear visibility at all times.

Following buzzer activation, participants return to the starting position and recall the previously memorized letter pairs. After a five-second break, the next trial begins. Participants complete eight trials per locomotion method before proceeding to the next method.

Throughout the experiment, a virtual display provides information about the currently active locomotion method and the respective task phase (see Figure 1). The custom software developed in Unity automatically logs whether participants activate the buzzer with the correct hand using the corresponding Oculus Quest controller and records the required time. Recalled letter pairs are documented by the experimenter.

3 Experiment

The study employed a 2×3 mixed factorial design with posture (*standing* vs. *sitting*) as a between-subjects factor and locomotion method (*Real*, *Joystick*, *Teleport*) as a within-subjects factor. Dependent variables included task performance, task duration, and subjective assessments of mental effort, stress, and preference.

3.1 Procedure

After providing informed consent, participants completed a demographic questionnaire and a tutorial to familiarize themselves with the tasks and locomotion methods. They then completed experimental trials using all three methods in counterbalanced order (Latin square design) within their assigned posture condition. The entire procedure took approximately 25 minutes.

3.2 Participants

A total of 42 participants (22 female, 20 male) took part in the study, with a mean age of 24.2 years ($SD = 2.9$). All participants were students or doctoral candidates and were evenly distributed across the two posture conditions (21 per condition), with balanced age and gender distribution. Participants reported moderate prior experience with VR, rated on a seven-point Likert scale (1 = never, 7 = expert; $M = 3.0$, $SD = 1.8$). Experience with the VR locomotion methods was assessed using the same scale and was comparable across methods, with mean ratings for *Real* ($M = 2.2$, $SD = 1.8$), *Joystick* ($M = 2.2$, $SD = 1.9$), and *Teleport* ($M = 1.7$, $SD = 1.7$).

4 Results

Prior to statistical analyses, assumptions were evaluated using the Shapiro–Wilk test for normality, Levene’s test for homogeneity of variances, and Mauchly’s test for sphericity (all $p > .05$). Unless indicated otherwise, data were analyzed using two-way mixed analyses of variance (ANOVAs) with posture (*standing* vs. *sitting*) as a between-subjects factor and locomotion method (*Real*, *Joystick*, *Teleport*) as a within-subjects factor. Bonferroni-corrected post-hoc tests were applied where appropriate, with the significance level set to $\alpha = .05$.

4.1 Task Performance

Letter Recall. A two-way mixed ANOVA was conducted on the mean proportion of correctly recalled letter pairs (see Figure 2). The locomotion method $\times$ posture interaction was not significant, $F(2, 80) = 0.57$, $p = .569$, $\eta_p^2 = .014$. A significant main effect of locomotion method was observed, $F(2, 80) = 4.54$, $p = .014$, $\eta_p^2 = .102$, whereas the main effect of posture was not significant, $F(1, 40) = 0.53$, $p = .472$, $\eta_p^2 = .013$. Bonferroni-corrected post-hoc comparisons revealed that participants recalled significantly more letter pairs in the *Joystick* condition than in the *Teleport* condition ($p = .014$).

Buzzer Hit Accuracy. The mean proportion of correctly activated buzzers is illustrated in Figure 2. Mauchly’s test indicated a violation of the sphericity assumption ($\epsilon = .645$); therefore, Greenhouse–Geisser corrected degrees of freedom are reported. The locomotion method $\times$ posture interaction was not significant, $F(1.29, 51.64) = 0.27$, $p = .664$, $\eta_p^2 = .007$. A significant main effect of locomotion method was observed, $F(1.29, 51.64) = 4.54$, $p = .029$, $\eta_p^2 = .102$. Bonferroni-adjusted post-hoc tests showed significantly higher accuracy in the *Real* condition than in the *Teleport* condition ($p = .015$). The main effect of posture was not significant, $F(1, 40) = 0.05$, $p = .825$, $\eta_p^2 = .001$.

4.2 Task Duration

Task duration, defined as the time from movement onset to successful buzzer activation, is shown in Figure 2. Levene’s test indicated heterogeneity of variances ($p = .004$); therefore, a Box–Cox transformation ($\lambda = -0.07$) was applied prior to analysis. A two-way mixed ANOVA revealed a significant locomotion method $\times$ posture interaction, $F(2, 80) = 10.68$, $p < .001$, $\eta_p^2 = .211$. Post-hoc analyses indicated that participants in the *Real* condition completed the task significantly faster when *standing* than when *sitting* ($p = .002$). A

significant main effect of locomotion method was also observed, $F(2, 80) = 98.67$, $p < .001$, $\eta_p^2 = .712$. Pairwise comparisons revealed significant differences between *Real* and *Joystick* ($p < .001$), *Real* and *Teleport* ($p = .014$), and *Joystick* and *Teleport* ($p < .001$). The main effect of posture was not significant, $F(1, 40) = 2.58$, $p = .116$, $\eta_p^2 = .061$.

4.3 Subjective Assessments

This subsection presents subjective measures of mental effort, stress, and participants’ preferences for locomotion. Mental effort and stress were assessed using seven-point Likert scales ranging from 1 (strongly disagree) to 7 (strongly agree). Preferences for locomotion methods were assessed using an ordinal forced-choice ranking, with unique ranks from 1 (least preferred) to 3 (most preferred).

Mental Effort. Perceived mental effort (see Figure 3) was assessed with the item “I required a high mental effort to solve the tasks”. The locomotion method $\times$ posture interaction was significant, $F(2, 80) = 5.50$, $p = .006$, $\eta_p^2 = .121$. There was no significant main effect of posture, $F(1, 40) = 0.02$, $p = .878$, $\eta_p^2 < .001$, but a significant main effect of locomotion method, $F(2, 80) = 12.69$, $p < .001$, $\eta_p^2 = .241$. Post-hoc comparisons revealed that the *Teleport* condition required significantly more mental effort than both the *Real* ($p = .003$) and *Joystick* ($p < .001$) conditions.

Stress Level. Perceived stress (see Figure 3) was assessed using the statement “While completing the respective locomotion tasks, I was very stressed”. The locomotion method $\times$ posture interaction was significant, $F(2, 80) = 5.45$, $p = .006$, $\eta_p^2 = .120$. A significant main effect of locomotion method was observed, $F(2, 80) = 18.82$, $p < .001$, $\eta_p^2 = .320$. Bonferroni-adjusted post-hoc tests indicated that stress levels were lowest in the *Joystick* condition compared to *Real* ($p = .028$) and *Teleport* ($p < .001$), and were also significantly lower in the *Real* condition than in the *Teleport* condition ($p = .009$). The main effect of posture was not significant, $F(1, 40) = 0.13$, $p = .718$, $\eta_p^2 = .003$.

Preferred Locomotion Method. Preferences for locomotion methods (see Figure 3) were analyzed using a cumulative link mixed model with locomotion method, posture, and their interaction as fixed effects, and a random intercept for participant. Likelihood-ratio tests revealed a significant locomotion method $\times$ posture interaction, $\chi^2(2) = 16.27$, $p < .001$, as well as significant main effects of locomotion method, $\chi^2(4) = 55.65$, $p < .001$, and posture, $\chi^2(3) = 16.33$, $p < .001$. Due to the significant interaction, locomotion method differences were interpreted using posture-specific post-hoc comparisons. Overall, averaged across postures, *Joystick* received the highest rankings, followed by *Real* and *Teleport*. When *standing*, *Joystick* and *Real* were ranked significantly higher than *Teleport* (both $p < .001$), with no difference between *Joystick* and *Real*. In the *sitting* condition, *Joystick* was ranked significantly higher than both *Real* ($p = .002$) and *Teleport* ($p < .001$), whereas *Real* and *Teleport* did not differ. Posture-specific comparisons further showed that *Real* was preferred when *standing* ($p = .005$), *Teleport* was preferred when *sitting* ($p = .012$), and *Joystick* was unaffected by posture ($p = .35$).

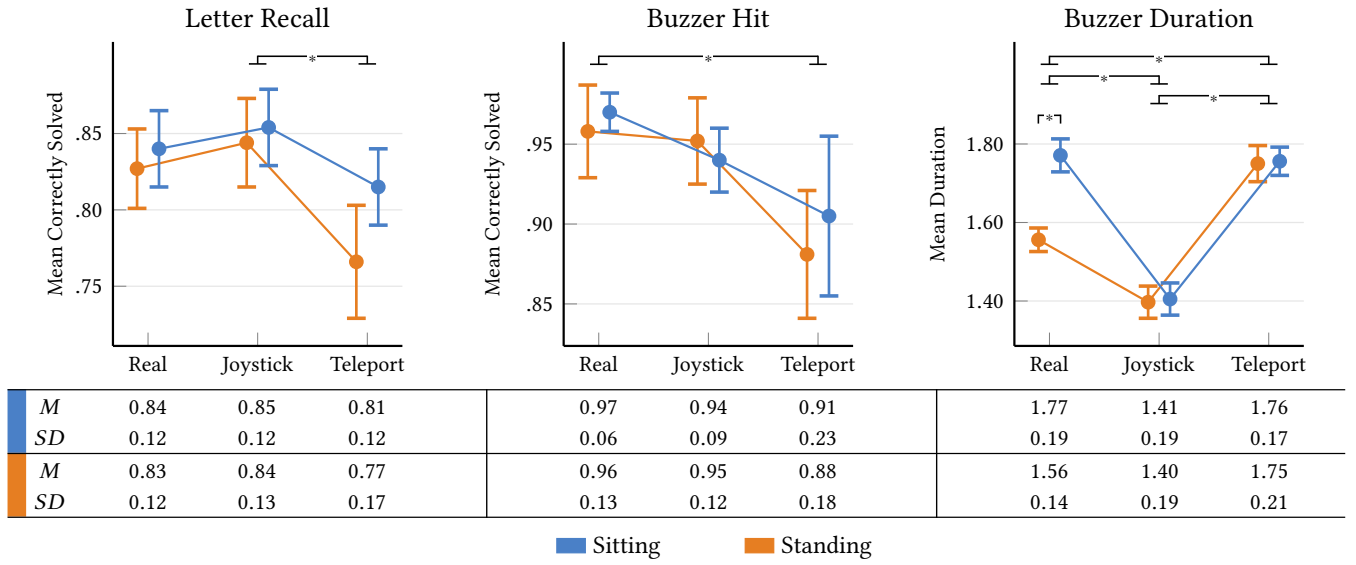


Figure 2: Performance measures across conditions. The panels report (from left to right) letter-recall accuracy, the proportion of correct buzzer activations, and the transformed buzzer-activation time. Error bars represent standard errors.

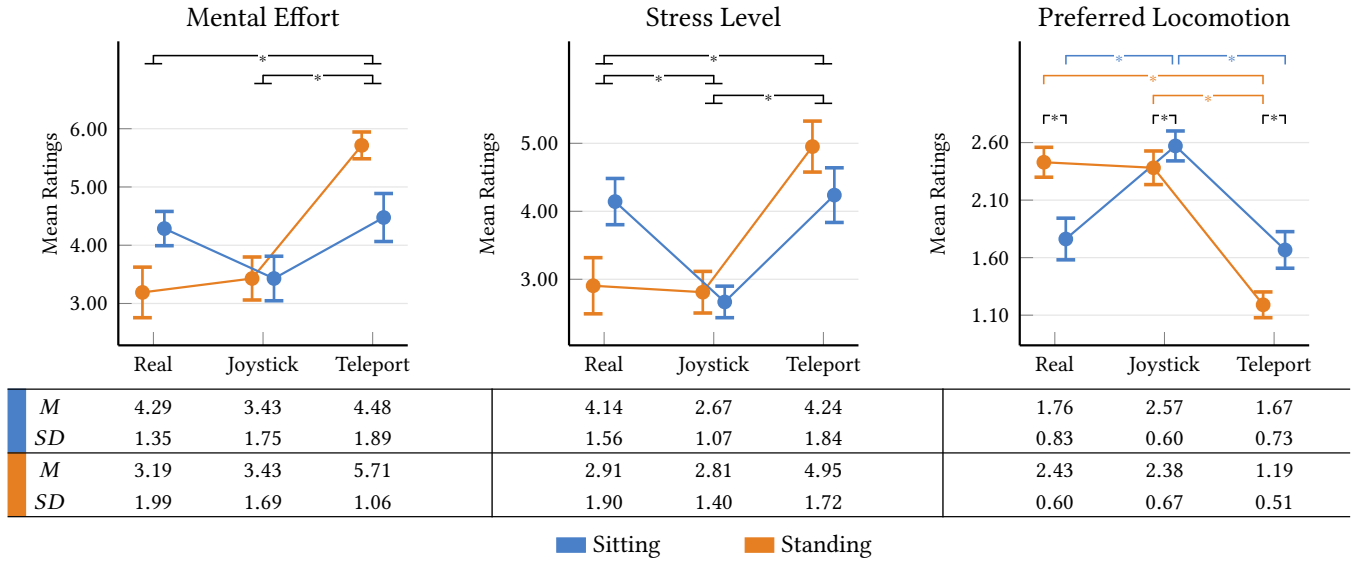


Figure 3: Subjective ratings across measures. The left, center, and right panels show self-reported mental effort, perceived stress during task execution, and preferred locomotion methods, respectively. Higher values indicate more positive ratings. Error bars denote standard errors.

5 Discussion

The choice of locomotion method substantially influenced both objective performance and subjective workload. *Joystick* consistently yielded the best outcomes, whereas *Teleport* with orientation control imposed the highest cognitive demands across postures. Buzzer hits served to increase task difficulty rather than assess cognitive ability, following prior work [13, 16, 17].

Our findings differ from those reported by Lai *et al.* [9]. In their study, the *Teleport* method required users to select only a target position, without specifying the resulting view orientation. In contrast, the *Teleport* method used in our study required participants to define both the target position and the view orientation, introducing an additional interaction step. This increased level of control may have contributed to the higher cognitive load observed in our *Teleport* condition.

The implementation of the *Joystick* method also differed from the setup used by Lai *et al.* In our study, participants adjusted their view via head movement as well as joystick input. Furthermore, locomotion speed was continuously scaled based on the displacement of the thumbstick. These additional control options may have contributed to the improved performance and reduced workload associated with joystick locomotion in our results. Moreover, when locomotion speed is not a critical factor, our results indicate that the *Real* method in standing posture offers a similarly effective alternative to *Joystick*.

Overall, these findings highlight that VR locomotion techniques differ considerably in their cognitive demands. They emphasize the importance of considering interaction complexity, including view control, target specification, and input modality, when evaluating locomotion methods. These results also underscore the need for further research to better understand how specific design features of locomotion techniques interact with cognitive load.

6 Conclusion

This study examined how three widely used VR locomotion methods—*Real*, *Joystick*, and *Teleport*—affect cognitive load under standing and sitting postures. Our results demonstrate that the choice of locomotion technique substantially influences both objective performance and subjective workload. *Joystick* consistently produced the fastest task completion times, the lowest perceived effort, and the highest user preferences across postures, whereas *Teleport* with orientation control imposed the highest cognitive demands. We observed posture-dependent effects, with *Real* performing worse in the seated condition. These findings highlight that implementation details such as orientation control and input modality play a critical role in shaping the cognitive demands of VR locomotion. We recommend that designers of VR applications carefully align locomotion techniques with intended postures and task requirements to optimize users' cognitive resources for learning- and training-relevant processing.

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